

Q1 - 25 July - Shift 1

Which of the following statements is a tautology ?

Space for your notes:

- (A) $((\sim p) \vee q) \Rightarrow p$ (B) $p \Rightarrow ((\sim p) \vee q)$
(C) $((\sim p) \vee q) \Rightarrow q$ (D) $q \Rightarrow ((\sim p) \vee q)$

Q2 - 25 July - Shift 2

Consider the following statements :

Space for your notes:

P : Ramu is intelligent

Q : Ramu is rich

R : Ramu is not honest

The negation of the statement "Ramu is intelligent and honest if and only if Ramu is not rich" can be expressed as :

- (A) $((P \wedge (\sim R)) \wedge Q) \wedge ((\sim Q) \wedge ((\sim P) \vee R))$
(B) $((P \wedge R) \wedge Q) \vee ((\sim Q) \wedge ((\sim P) \vee (\sim R)))$
(C) $((P \wedge R) \wedge Q) \wedge ((\sim Q) \wedge ((\sim P) \vee (\sim R)))$
(D) $((P \wedge (\sim R)) \wedge Q) \vee ((\sim Q) \wedge ((\sim P) \vee R))$

Q3 - 26 July - Shift 1

The statement $(\sim(p \Leftrightarrow \sim q)) \wedge q$ is :

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- (A) a tautology
(B) a contradiction
(C) equivalent to $(p \Rightarrow q) \wedge q$
(D) equivalent to $(p \Rightarrow q) \wedge p$

Q4 - 26 July - Shift 2

Questions

MathonGo

Negation of the Boolean expression $p \Leftrightarrow (q \Rightarrow p)$ is

- (A) $(\sim p) \wedge q$ (B) $p \wedge (\sim q)$
 (C) $(\sim p) \vee (\sim q)$ (D) $(\sim p) \wedge (\sim q)$

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Q5 - 27 July - Shift 1

 $(p \wedge r) \Leftrightarrow (p \wedge (\sim q))$ is equivalent to $(\sim p)$

when r is

- (A) p (B) $\sim p$
 (C) q (D) $\sim q$

Space for your notes:

Q6 - 27 July - Shift 2

If the truth value of the statement

 $(P \wedge (\sim R)) \rightarrow ((\sim R) \wedge Q)$ is F, then the truth value of which of the following is F ?

- (A) $P \vee Q \rightarrow \sim R$ (B) $R \vee Q \rightarrow \sim P$
 (C) $\sim(P \vee Q) \rightarrow \sim R$ (D) $\sim(R \vee Q) \rightarrow \sim P$

Space for your notes:

Q7 - 28 July - Shift 1

Let the operations $*, \odot \in \{\wedge, \vee\}$. If
 $(p * q) \odot (p \odot \sim q)$ is a tautology, then the ordered pair $(*, \odot)$ is :

- (A) $(\vee, \wedge)$ (B) $(\vee, \vee)$ (C) $(\wedge, \wedge)$ (D) $(\wedge, \vee)$

Space for your notes:

Q8 - 28 July - Shift 2

#MathBoleTohMathonGo

Questions

MathonGo

Let
 p : Ramesh listens to music.
 q : Ramesh is out of his village
 r : It is Sunday
 s : It is Saturday
 Then the statement “Ramesh listens to music only if he is in his village and it is Sunday or Saturday” can be expressed as

- (A) $((\sim q) \wedge (r \vee s)) \Rightarrow p$
 (B) $(q \wedge (r \vee s)) \Rightarrow p$
 (C) $p \Rightarrow (q \wedge (r \vee s))$
 (D) $p \Rightarrow ((\sim q) \wedge (r \vee s))$

Space for your notes:

Q9 - 29 July - Shift 1

The statement $(p \wedge q) \Rightarrow (p \wedge r)$ is equivalent to :

- (A) $q \Rightarrow (p \wedge r)$ (B) $p \Rightarrow (p \wedge r)$
 (C) $(p \wedge r) \Rightarrow (p \wedge q)$ (D) $(p \wedge q) \Rightarrow r$

Space for your notes:

Q10 - 29 July - Shift 2

The statement $(p \Rightarrow q) \vee (p \Rightarrow r)$ is NOT equivalent to:

- (A) $(p \wedge (\sim r)) \Rightarrow q$ (B) $(\sim q) \Rightarrow ((\sim r) \vee p)$
 (C) $p \Rightarrow (q \vee r)$ (D) $(p \wedge (\sim q)) \Rightarrow r$

Space for your notes:

Answer Key

Q1 (D)

Q2 (D)

Q3 (D)

Q4 (D)

Q5 (C)

Q6 (D)

Q7 (B)

Q8 (D)

Q9 (D)

Q10 (B)

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Q1 (D)

p	q	$\sim p$	$\sim q$	$\sim p \vee q$
T	T	F	F	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

options

1	2	3	4
T	T	T	T
T	F	T	T
F	T	T	T
F	T	F	T

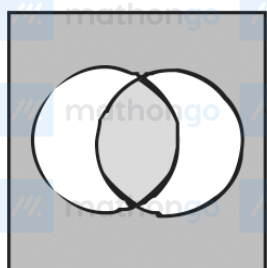
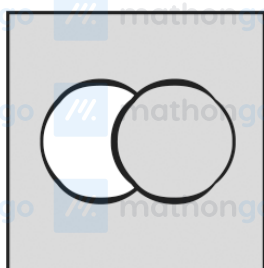
Q2 (D)

(D)

Q3 (D)

$$(\sim(p \Leftrightarrow \sim q)) \wedge q \equiv (p \Leftrightarrow q) \wedge q$$

$$(p \Leftrightarrow q) \wedge q \equiv p \wedge q$$

 $p \Leftrightarrow q$  $p \rightarrow q$

Q4 (D)

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$$\sim(p \leftrightarrow (q \rightarrow p))$$

$$\sim(p \leftrightarrow q) = (p \wedge \sim q) \vee (q \wedge \sim p)$$

$$\sim(p \leftrightarrow (q \rightarrow p)) = (p \wedge \sim(q \rightarrow p)) \vee ((q \rightarrow p) \wedge \sim p)$$

$$(p \wedge \sim(q \rightarrow p)) = p \wedge (q \wedge \sim p) = (p \wedge \sim p) \wedge q = c$$

$$(q \rightarrow p) \wedge \sim p = (\sim q \vee p) \wedge \sim p = \sim p \wedge (\sim q \vee p)$$

$$= (\sim p \wedge \sim q) \vee (\sim p \wedge p) = \sim p \wedge \sim q$$

$$\sim(p \leftrightarrow (q \rightarrow p)) = c \vee (\sim p \wedge \sim q) = \sim p \wedge \sim q$$

Q5 (C)

Given $(p \wedge r) \Leftrightarrow (p \wedge (\sim q)) \equiv (\sim p)$

Taking $r = q$

p	q	$\sim p$	$\sim q$	$p \wedge q$	$p \wedge \sim q$	$(p \wedge r) \Leftrightarrow (p \wedge (\sim q))$
T	T	F	F	T	F	F
T	F	F	T	F	T	F
F	T	T	F	F	F	T
F	F	T	T	F	F	T

So, clear $(p \wedge r) \Leftrightarrow (p \wedge (\sim q)) \equiv (\sim p)$

Q6 (D)

$X \Rightarrow Y$ is a false
when X is true and Y is false

So, $P \rightarrow T, Q \rightarrow F, R \rightarrow F$

(A) $P \vee Q \rightarrow \sim R$ is T

(B) $R \vee Q \rightarrow \sim P$ is T

(C) $\sim (P \vee Q) \rightarrow \sim R$ is T

(D) $\sim (R \vee Q) \rightarrow \sim P$ is F

Q7 (B)

Well check each option

For A $\pi = v$ of $0 = \Lambda$

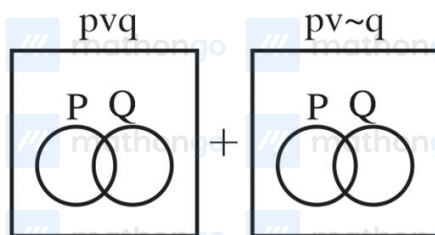
$$(p \vee q) \wedge (p \vee \sim q)$$

$$\equiv p \vee (q \wedge \sim q)$$

$$\equiv p \vee (c) \equiv p$$

For B : $* = v, O = v$

$$(p \vee q) \vee (p \vee \sim q) \equiv t \quad \text{using Venn Diagrams}$$



#MathBoleTohMathonGo

Q8 (D)

 $p \equiv$ Ramesh listens to music $\sim q \equiv$ He is in village. $r \vee s \equiv$ Saturday or sunday $p \Rightarrow ((\sim q) \wedge (r \vee s))$

Q9 (D)

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$$(p \wedge q) \Rightarrow (p \wedge r)$$

$$\sim(p \wedge q) \vee (p \wedge r)$$

$$(\sim p \vee \sim q) \vee (p \wedge r)$$

$$(\sim p \vee (p \wedge r)) \vee \sim q$$

$$(\sim p \vee p) \wedge (\sim p \vee r) \vee \sim q$$

$$(\sim p \vee r) \vee \sim q$$

$$(\sim p \vee \sim q) \vee r$$

$$\sim(p \wedge q) \vee r$$

$$(p \wedge q) \Rightarrow r$$

Q10 (B)

$$(p \rightarrow q) \vee (p \rightarrow r)$$

$$(\sim p \vee q) \vee (\sim p \vee r)$$

$$= \sim p \vee (q \vee r)$$

$$= p \rightarrow (q \vee r) \equiv (3) \text{ is true.}$$

Now (1) $(p \wedge \sim r) \rightarrow q$

$$\sim(p \wedge \sim r) \vee q = (\sim p \vee r) \vee q$$

$$= \sim p \vee (r \vee q) = p \rightarrow (q \vee r)$$

$$(4) (p \wedge \sim q) \rightarrow r = p \rightarrow (q \vee r)$$